DATA ANALYTICS WITH POWER BI

PROJECT REPORT

(Project Semester January-April 2025)



Cross-Platform Engagement Dynamics: Reddit Discussions and Google Search Interest

Submitted by

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Course Code : INT374

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CERTIFICATE

This is to certify that Madhusmita Talukdar bearing Registration no. 12314142 has completed INT374 project titled, **"Cross-Platform Engagement Dynamics: Reddit Discussions and Google Search Interest"** under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

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Date: 20-12-2025

DECLARATION

I, Madhusmita Talukdar, student of Bachelor of Technology under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 20-12-2025 Signature

Registration No. 12314142 Madhusmita Talukdar

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to all those who supported and guided me throughout the completion of this project titled

"Cross-Platform Engagement Dynamics: Reddit Discussions and Google Search Interest"

First and foremost, I would like to extend my heartfelt thanks to my respected mentor and Power BI instructor for their invaluable guidance, continuous support, and encouragement throughout the development of this project. Their expert insights, practical approach to data analytics, and timely feedback played a crucial role in shaping my understanding and helped me successfully design and complete this dashboard.

I would also like to express my appreciation to the faculty and staff of my institution for providing a supportive academic environment and the necessary resources that facilitated this learning experience.

A special note of thanks goes to my peers, friends, and family members for their constant motivation, encouragement, and moral support during the course of this project.

Lastly, I would like to acknowledge the use of publicly available financial market data and news sources, including Yahoo Finance RSS feeds, which served as the foundation for the stock price analysis, volume insights, and sentiment analysis performed in this project.

Madhusmita Talukdar

12314142

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1. INTRODUCTION

1.1 Overview of Business Intelligence & Analytics

Business Intelligence (BI) focuses on transforming raw data into meaningful insights using data modeling, visualization, and analytical techniques. Power BI enables interactive dashboards that support decision-making through visual storytelling rather than predictions alone.

1.2 Importance of Analytics in Social Media & Public Interest

Online platforms such as Reddit reflect real user engagement, while Google Trends captures public search demand. Analyzing these together helps identify:

- Content demand gaps
- Oversaturated domains
- High-engagement discussion areas
- Time-based engagement behavior

1.3 Problem Statement

There is a lack of integrated dashboards that combine:

- Social media engagement (Reddit)
- Public interest signals (Google Trends)

to evaluate demand vs supply across content domains.

1.4 Objectives of the Project

- Analyze engagement quantity and quality across domains
- Measure virality and discussion depth
- Compare Reddit engagement with Google search demand
- Identify underserved and oversaturated domains
- Provide executive-level insights using Power BI

1.5 Scope of the Project

- Multi-domain analysis (Sports, Technology, Healthcare, Politics, etc.)
- Time-based analytics (daily, monthly, hourly)
- Demand–supply intelligence
- Interactive Power BI dashboards

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2. SOURCE OF DATASET

2.1 Datasets Used

Dataset	Source
Reddit Multi-Domain Engagement	**Reddit API**
Public Interest Data	**Google Trends**

2.2 Dataset Description

Reddit Dataset

- PostID
- Domain
- Subreddit
- Score
- Comments
- Upvote Ratio
- Engagement Bucket
- Created Date
- NSFW Flag

Google Trends Dataset

- Domain
- Date
- Average Search Interest
- Trend Score

2.3 Dataset Size

- Total Records: 9,000
- Domains Covered: 10
- Time Period: June – December

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3. DATA PREPROCESSING & MODELLING (POWER BI)

3.1 Data Cleaning

- Removed null values
- Standardized domain names
- Filtered irrelevant records

3.2 Data Modeling

A star schema was implemented:

- Fact Tables
    - Reddit Engagement
    - Google Trends
- Dimension Tables
    - Date Dimension
    - Domain Dimension

3.3 Calculated Measures (DAX)

Examples:

- Total Engagement
- Average Engagement per Post
- Viral Posts %
- Engagement Buckets
- Demand Index
- Supply Index
- Demand–Supply Index
- Market Signal (Underserved / Balanced / Oversaturated)

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4. DASHBOARD & ANALYSIS

Page 1: Overview

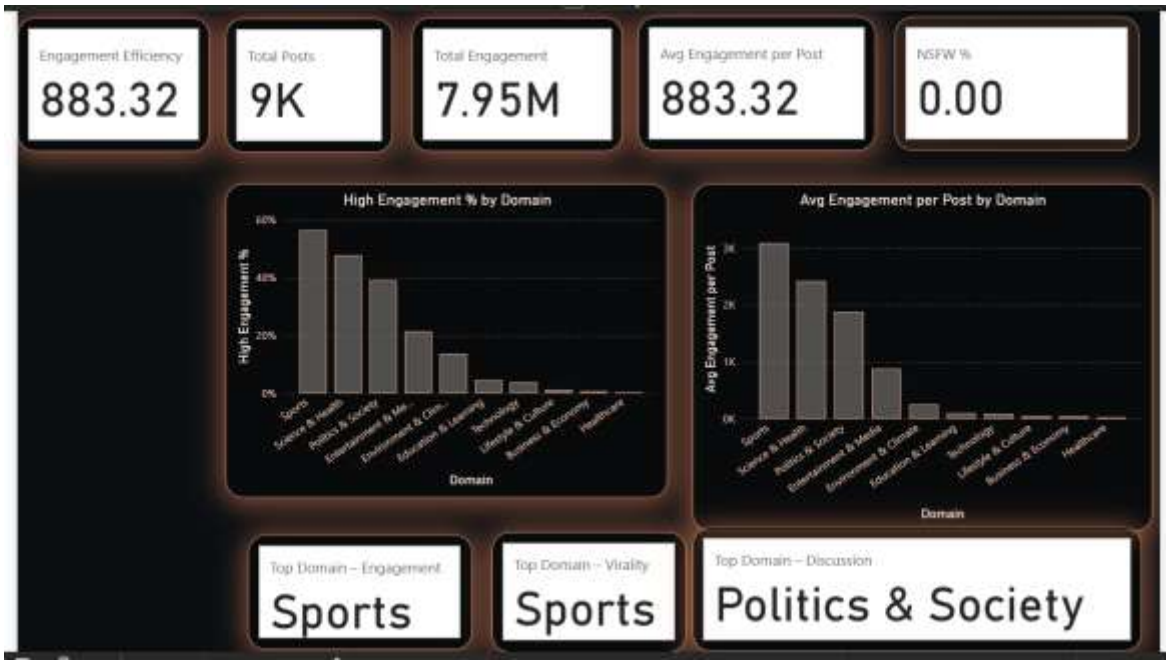
Purpose: High-level summary for stakeholders

Key Visuals

- KPI Cards (Total Posts, Total Engagement, Engagement Efficiency)
- Top Domain – Engagement, Virality, Discussion
- Overall engagement distribution

Insight

- Sports dominates engagement and virality
- Politics & Society drives deeper discussions



Page 2: Executive Overview

Purpose: Decision-ready insights

Key Visuals

- Top 3 Domains (Text Insight Cards)
- High Engagement % by Domain
- Avg Engagement per Post

Insight

- Sports: High engagement & virality
- Science & Health: High engagement efficiency
- Politics & Society: Strong discussion depth

Page 3: Domain Performance

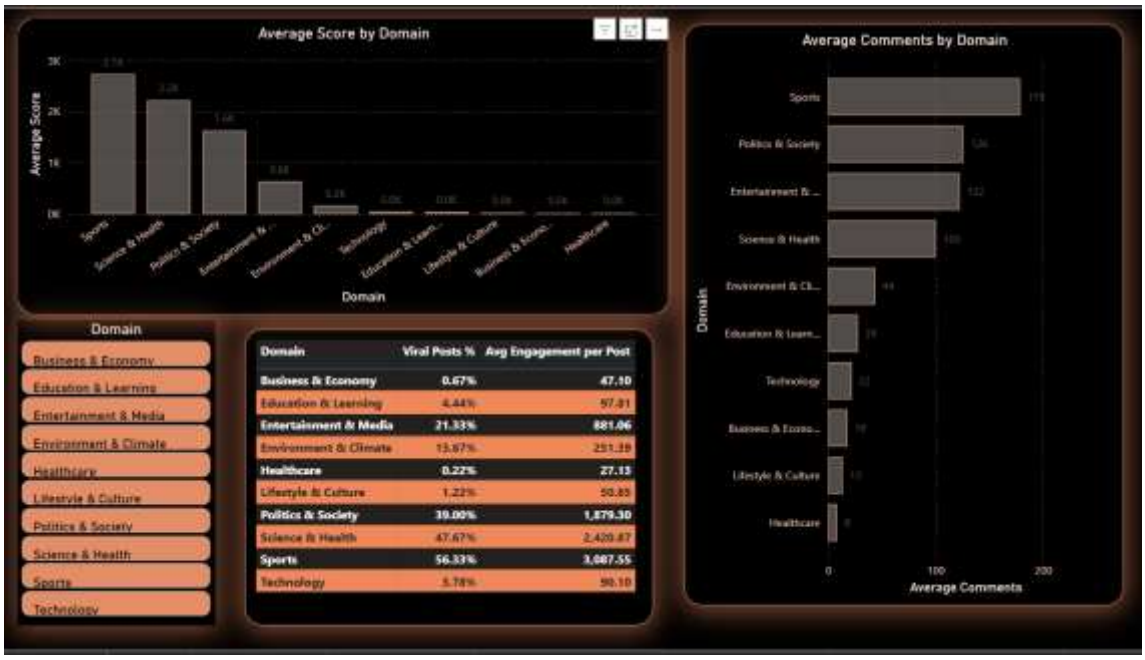
Purpose: Compare domains on engagement metrics

Key Visuals

- Average Score by Domain
- Average Comments by Domain
- Total Posts vs Avg Engagement

Insight

- High volume does not always mean high engagement
- Some domains are content-heavy but engagement-poor



Page 4: Engagement Quality & Virality

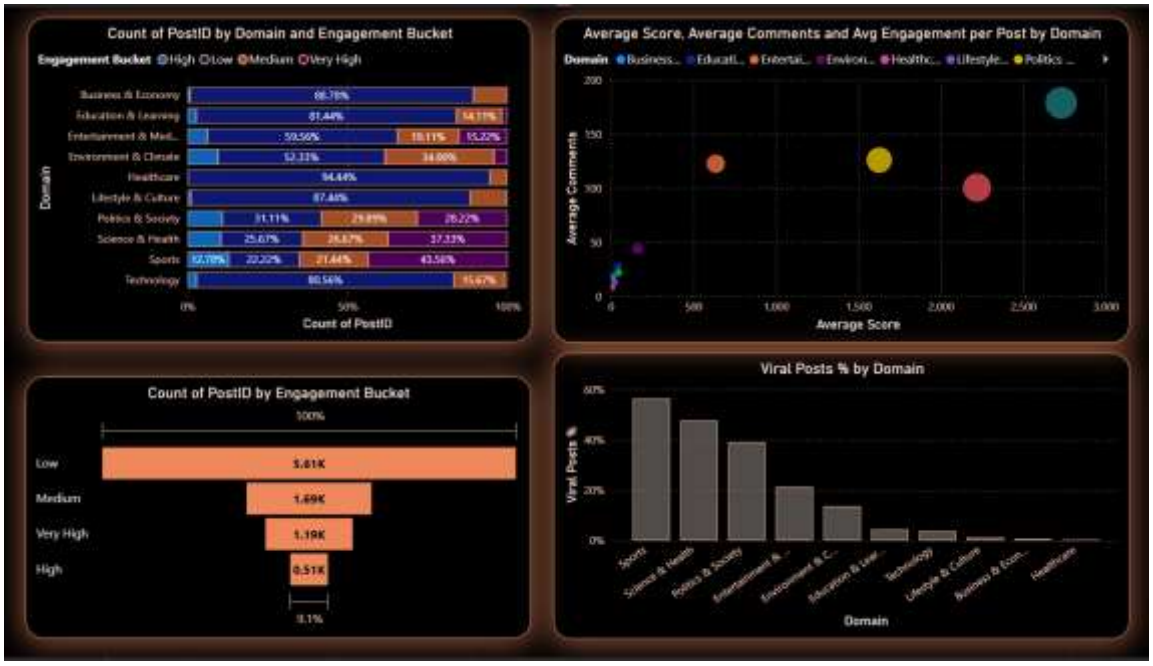
Purpose: Understand content quality

Key Visuals

- Engagement Bucket Distribution
- Viral Posts %
- Engagement Funnel

Insight

- Only ~9% posts reach high or very high engagement
- Sports and Science & Health dominate viral content



Page 5: Time Intelligence Analysis

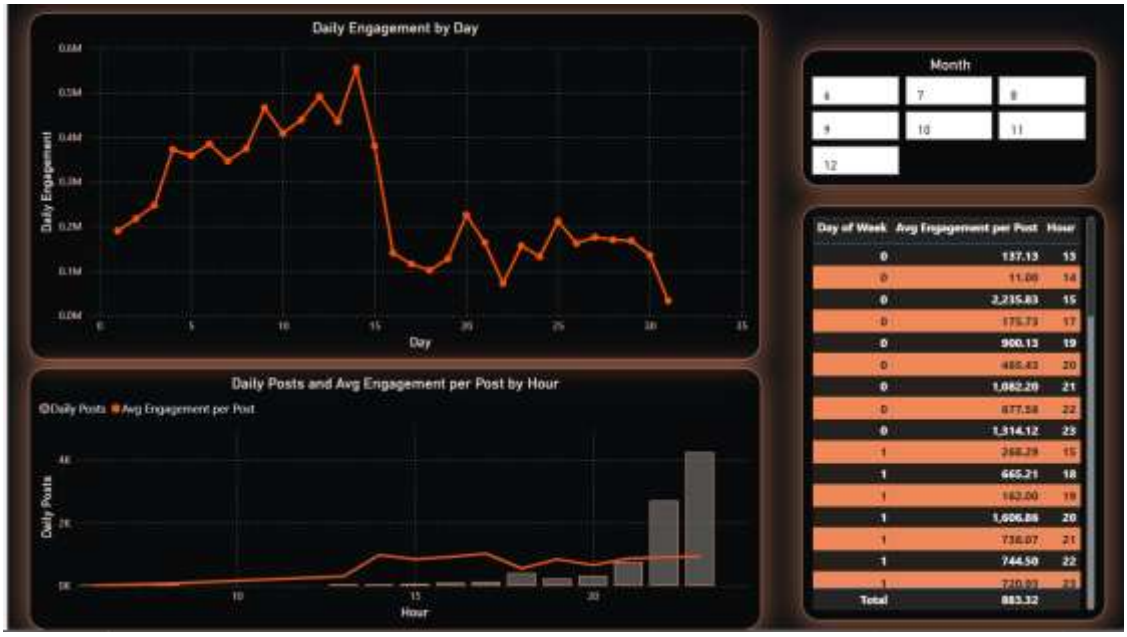
Purpose: Identify best posting times

Key Visuals

- Daily Engagement by Day
- Engagement by Hour
- Monthly Engagement Trends

Insight

- Engagement peaks in November–December
- Late evening hours (8-11) show maximum engagement



Page 6: Public Interest (Google Trends Analysis)

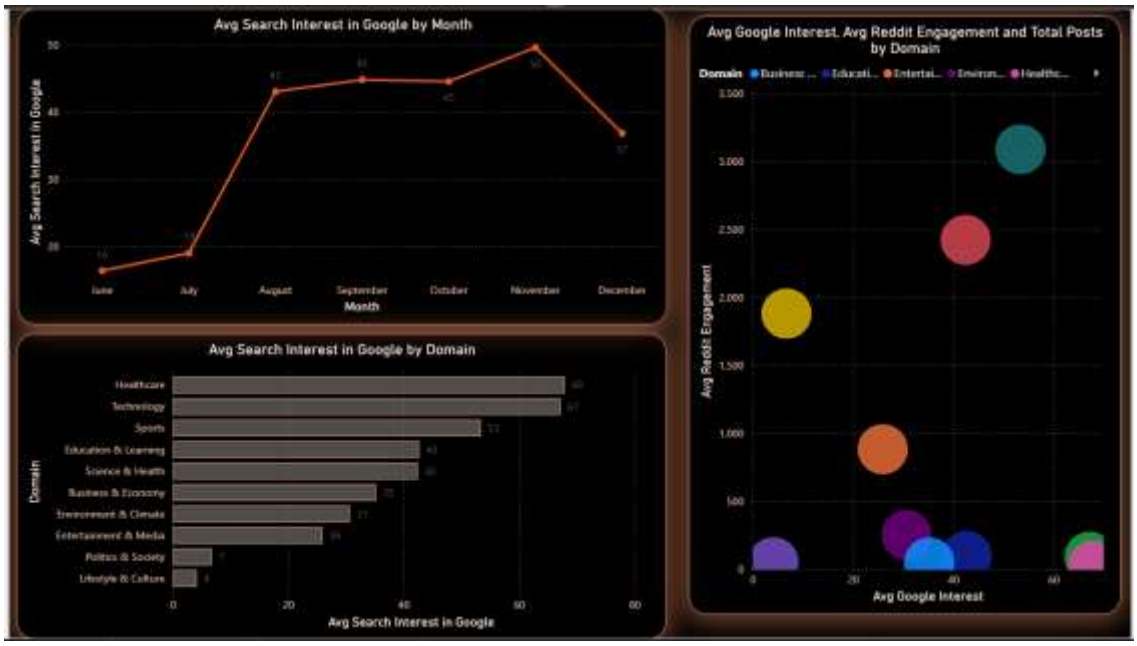
Purpose: Analyze search demand

Key Visuals

- Avg Google Interest by Month
- Avg Google Interest by Domain
- Bubble chart: Google Interest vs Reddit Engagement

Insight

- Healthcare & Technology show high public interest
- Some high-interest domains have lower Reddit supply



Page 7: Demand vs Supply (Strategic Insights)

Purpose: Identify opportunities

Key Visuals

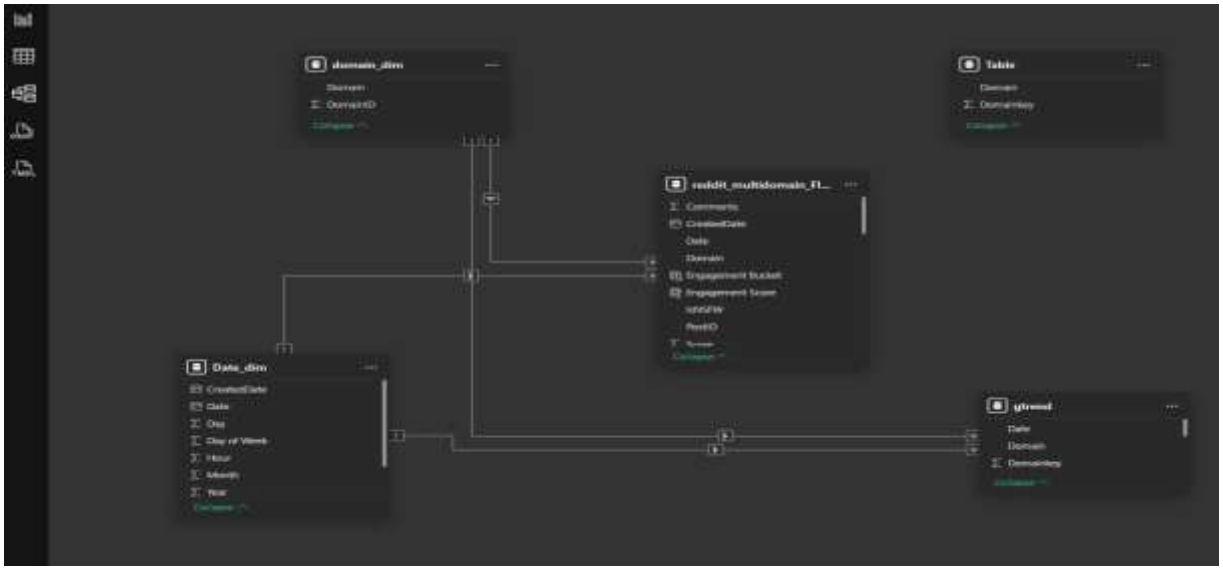
- Demand–Supply Index by Domain
- Market Signal Table
- Underserved vs Oversaturated Domains

Insight

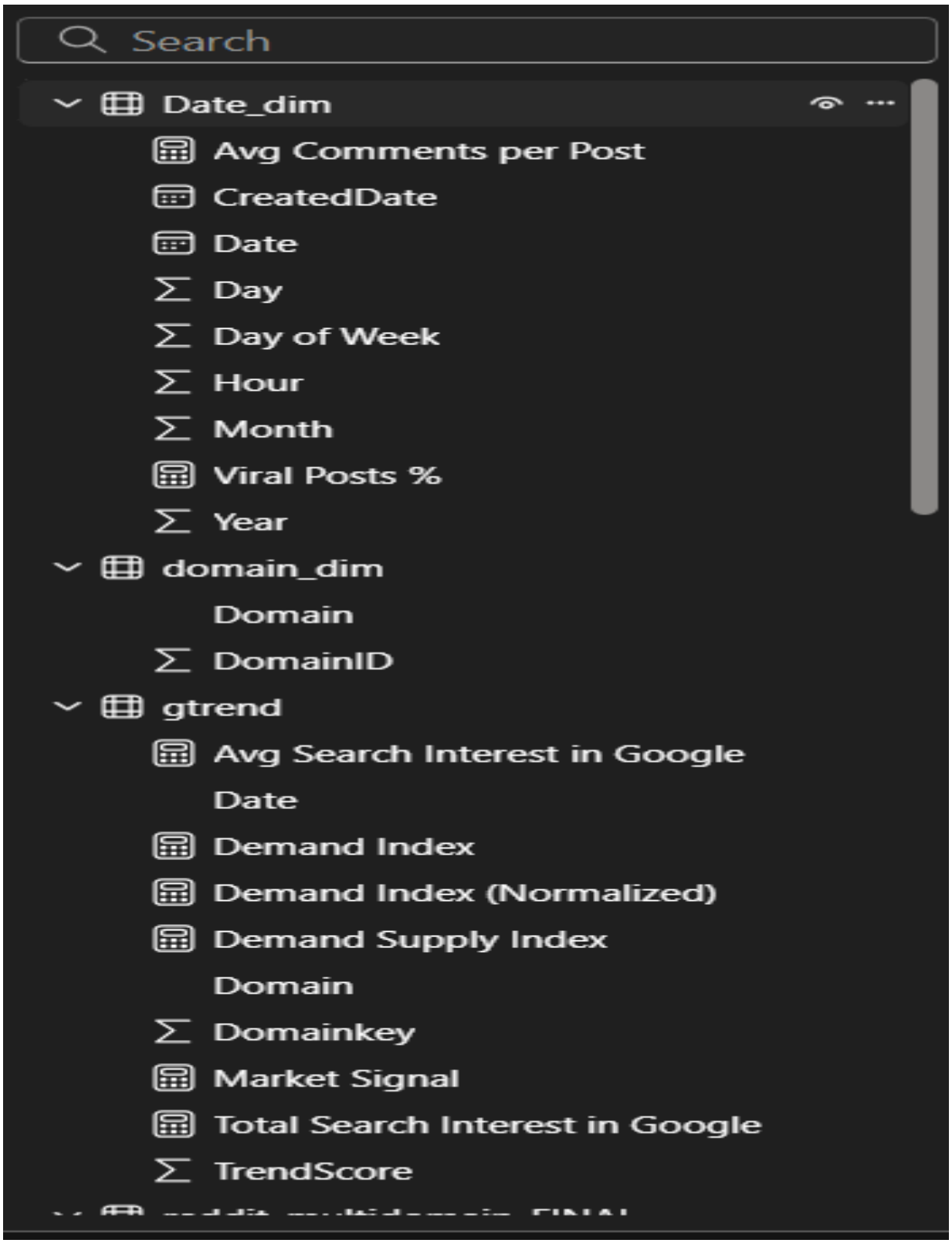
- Healthcare & Technology are underserved
- Entertainment shows oversupply
- Strong opportunities for content creation in high-demand, low-supply domains

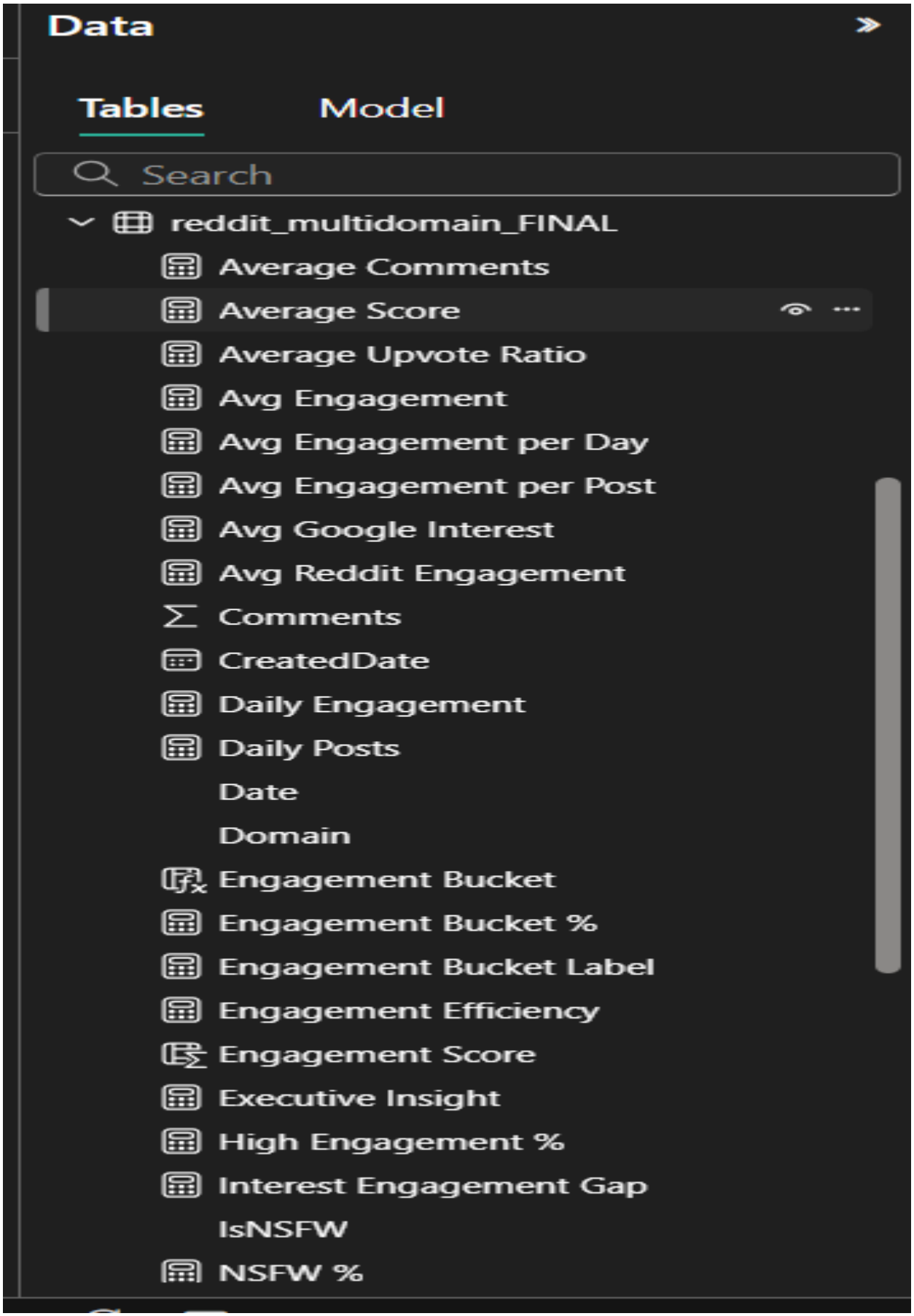


Data Modelling



MEASURES AND OTHER COLUMNS




Data
Tables
Model
Search
reddit_multidomain_FINAL
Average Comments
Average Score
Average Upvote Ratio
Avg Engagement
Avg Engagement per Day
Avg Engagement per Post
Avg Google Interest
Avg Reddit Engagement
Comments
CreatedDate
Daily Engagement
Daily Posts
Date
Domain
Engagement Bucket
Engagement Bucket %
Engagement Bucket Label
Engagement Efficiency
Engagement Score
Executive Insight
High Engagement %
Interest Engagement Gap
IsNSFW
NSFW %

5. KEY FINDINGS

- Engagement and demand are not always aligned
- Sports dominates engagement and virality
- Healthcare & Technology show high demand but lower content supply
- Late evenings and end-year months yield higher engagement

6. CONCLUSION

This Power BI project successfully demonstrates how multi-source data analytics can uncover meaningful demand–supply insights. By integrating Reddit engagement with Google Trends, the dashboard enables data-driven content and strategy decisions**.**

7. FUTURE SCOPE

- Real-time API integration
- Sentiment analysis on Reddit comments
- Geographic demand comparison
- Automated alerts for demand spikes

8. REFERENCES

- Reddit API Documentation
- Google Trends Documentation
- Microsoft Power BI Documentation
- Dataset : The dataset used in this project was collected dynamically using public APIs rather than static files.

  • Reddit Engagement Data: Retrieved using the Reddit API, which provides access to post metadata such as scores, comments, upvote ratios, and timestamps across multiple subreddits.

  • Public Interest Data: Retrieved from Google Trends using domain-level keyword queries to measure public search interest over time.

  Since the data was fetched programmatically via APIs and processed in Power BI, no single static dataset link exists.(https://www.reddit.com/dev/api/ , https://trends.google.com/trends/)
- Github : https://github.com/Madhusmita111/POWER-BI-DASHBOARD
- LinkedIn : https://www.linkedin.com/in/madhusmitatalukdar/

- Google Drive : https://drive.google.com/drive/u/1/folders/1akCfsiZwchhcnWZS8EOXQltswSEMnUeY